BUSINESS BLUEPRINT

AOP AUTOMATION FOR DMD VCM PLANT

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Business Blueprint

AOP Automation for DMD VCM Plant

Prepared for:

Reliance Industries Ltd

Submitted by:

Honeywell Automation India Ltd (“Honeywell”)

Revision History

Rev.	Date	Originated By	Description of status and changes
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1. Introduction

The purpose of this document is to capture the detailed specifications for the Digitalized Annual Operating Plan (AOP) for VCM plant for DMD site.

This document is based on the Technical Proposal submitted by Honeywell to Reliance. It also refers to data collected during the workshop conducted at RCP with various VCM plant users across RIL sites.

1.1 Executive Summary

Reliance Industries Limited is India's leading private sector organization operating in various business verticals. The manufacturing sector is one such business vertical. RIL owns and operates multiple plants across various sites.

Defining and finalizing the Annual Operating Plan is a critical annual business process carried out by different functional teams. At present, this activity is carried out manually.

A Rev-0 budget is usually completed in Nov / Dec for the succeeding Financial Year (FY) and reviewed. The comments are addressed with updated information from the Business, as the next revised budget is expected by the end of January. Further revisions (if any) are incorporated in Feb / March to reflect updated business input as applicable.

The entries from the final rev of the budget are finalized in BPC before the beginning of the succeeding FY.

RIL aims to automate the AOP preparation process using a Data-Driven Model. This model leverages the last five years of AOP data from MIIS or SAP and incorporates plant-specific business rules (such as plant availability, running hours, feedstock availability, seasonal impact on production, etc.) to derive AOP figures in the expected formats.

As part of digitalization, Honeywell will support building a platform that will automate the Annual Operations Planning process for VCM technology.

1.2 Reference Documents

Document ID	Revision	Title	Document
DMD VCM DCT	-	Digital AOP DMD VCM DCT	DMD VCM EDC DCT Rev 01.xlsx

Table 1 Reference Documents

2. AOP Design Basis

The table below captures different functions for the scope of AOP digitization design –

Sr. No.	Functions	Covered in Document
1.	Health, Safety and Environment (HSE)	For later release, post review and discussion
2.	Process/Operations	Covered
3.	Reliability and Maintenance	For later release, post review and discussion

Table 2 AOP Functional Scope

Function-wise, the detailed design basis is captured in the sections below.

2.1 Process and Operations Function

In this section, the budgeted Production and Norms for a plant are derived, using a standard approach as briefed in the sections below for the following criteria

- Budget Production/Production AOP
- Norms/Consumption AOP

Steps to generate the Production and Consumption AOP are captured in the below flow diagram.

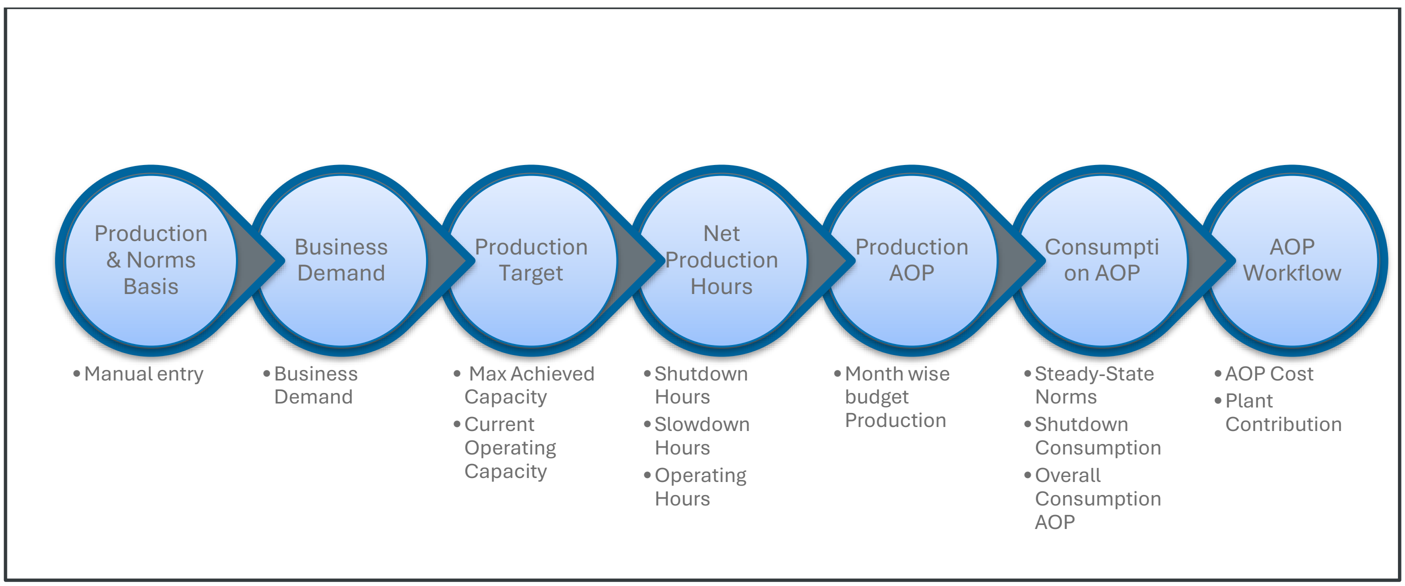


Figure 1 Digitized AOP Flow Diagram

2.1.1 Budget Production/Production AOP

Below are the steps captured to derive the budgeted AOP production values.

2.1.1.1 Production and Norms Basis

The Production and Norms Basis screen serves as the foundation for the digital AOP process, allowing users to input essential data for production and consumption AOP calculations.

AOP Design Basis serves as manual entry screen for Highlighting the key insights and summarizing the assumptions using a standard text box feature.

AOP historical period basis added where Start and End date is provided to select historical period to calculate Production Target data and Steady state norms for selected period by default 05 years data will be loaded.

The screenshot below highlights the details captured in the digitized AOP platform.

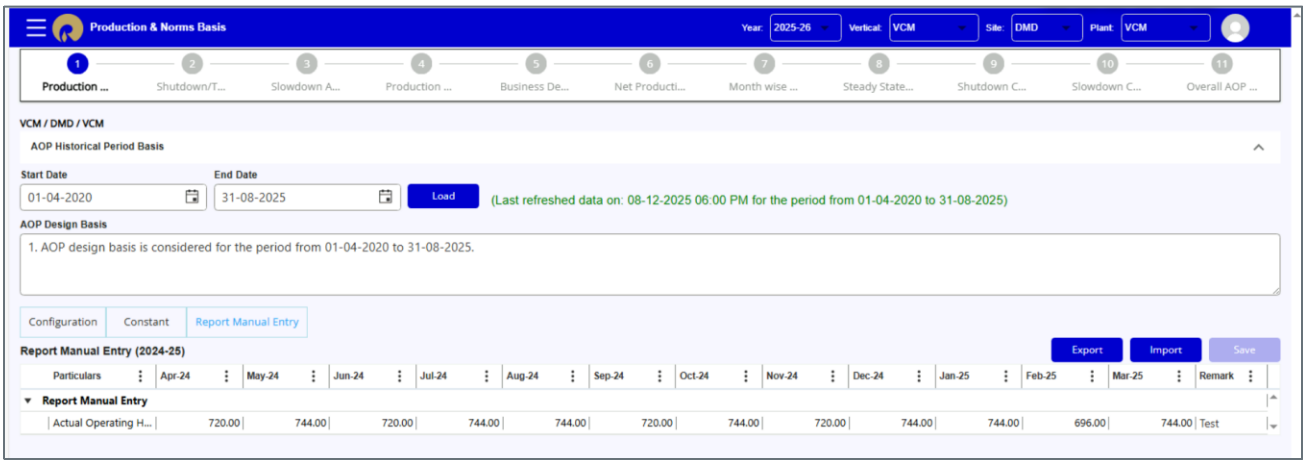


Figure 2 Digitized AOP screen for Production and Norms Basis

2.1.1.2 Business Demand

The production demand is considered as the maximum value from last year's production values achieved. The default value for VCM is 100%.

The business demand value can be modified by Business group upon providing a specific reason.

AOP digitization portal as seen in the screenshot below.

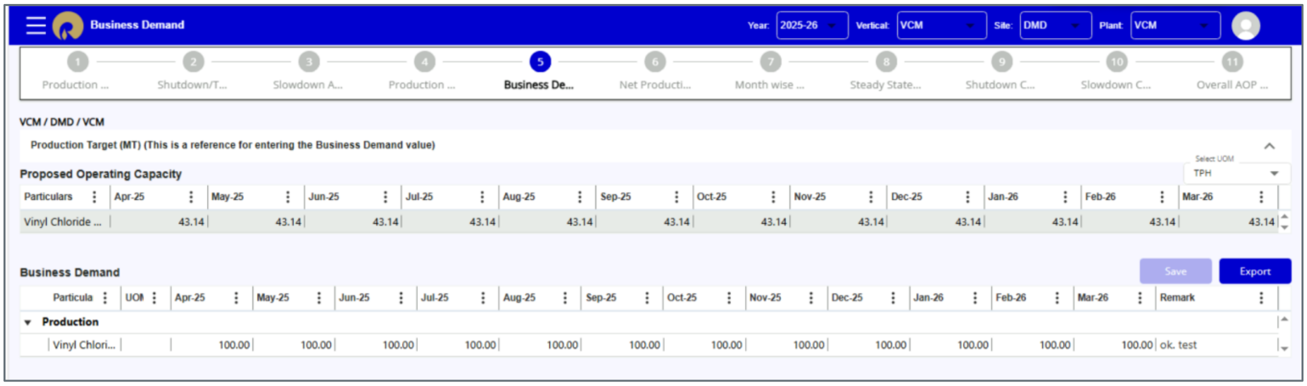


Figure 3 Digitized AOP screen for Business Demand

2.1.1.3 Production Target

Production target screen consists of below grid views.

a) Design Capacity:

Design Capacity is the plant capacity for different furnace modes of operation as per licensor provided data.

b) Max Achieved Capacity:

Max Achieved Capacity is derived as 01 month best from SAP Historical records.

c) Proposed Operating Capacity:

Proposed Operating Capacity will show data similar to Max Achieved Capacity. If the user needs to modify the VCM production rate this can be done by providing a specific reason in remarks column.

This information is captured in the AOP digitization portal as seen in the screenshot below.

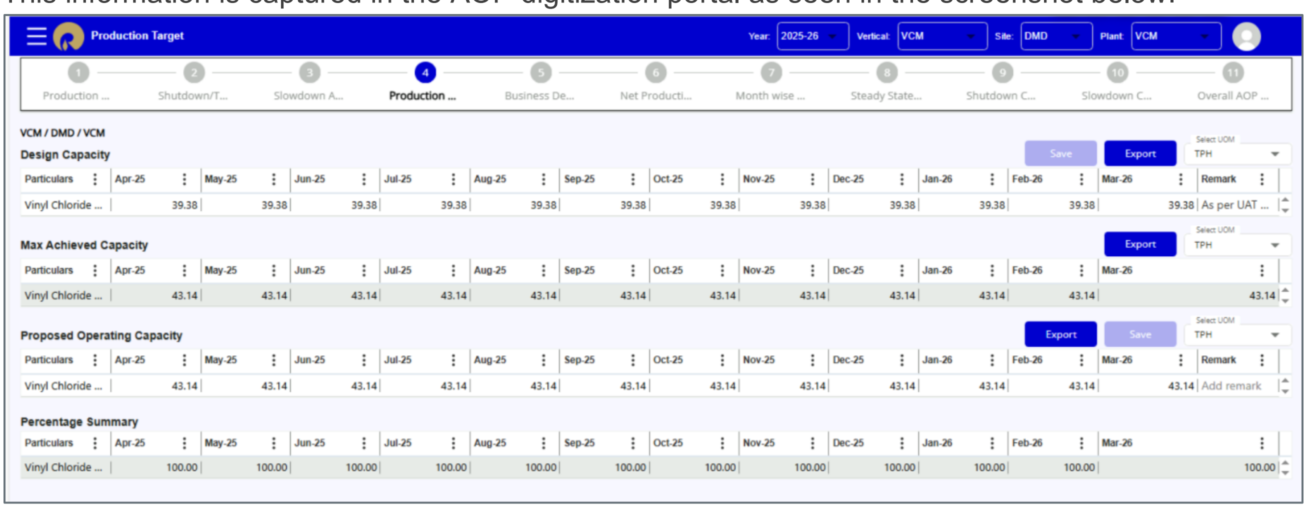


Figure 4 Digitized AOP screen for Production Target

2.1.1.4 Shutdown, Slowdown Activities and Net Production Hours

The operations team will provide the shutdown, turnaround and slowdown plan to calculate month-wise operating hours available for production.

The screenshot below depicts the shutdown and turnaround activities planned, as will be captured in the AOP digitized platform.

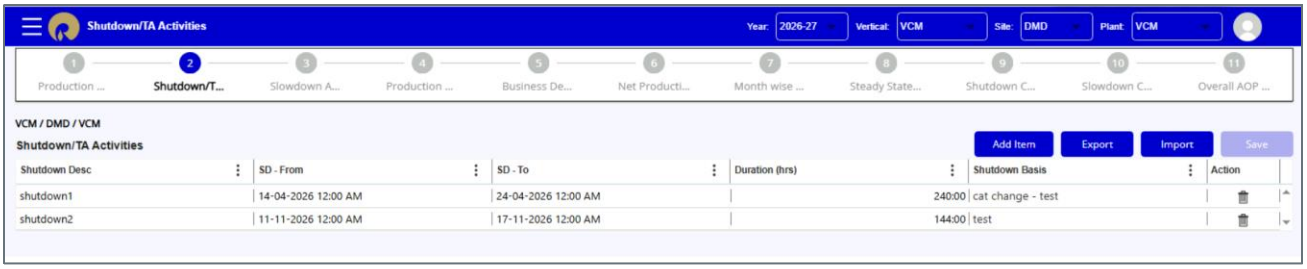


Figure 5 Digitized AOP screen for Shutdown/TA Activities

Following are the default options available in slowdown activities screen.

1) Furnace Decoking
2) Old Oxy Catalyst replacement
3) New Oxy Catalyst replacement
4) Plant Startup/Shutdown

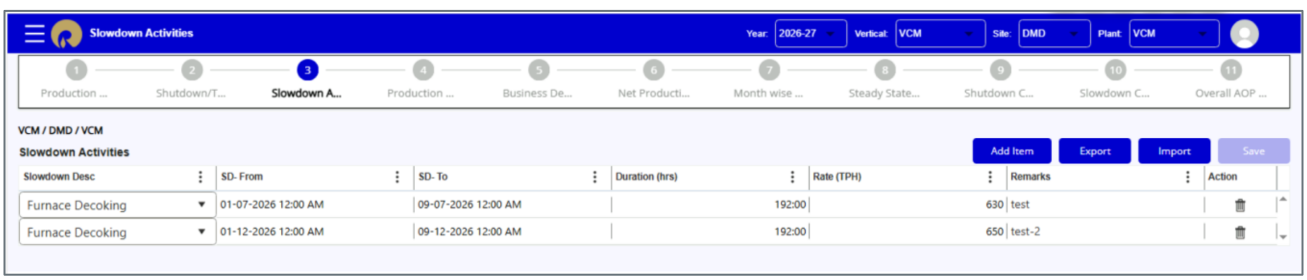


Figure 6 Digitized AOP screen for Slowdown Activities

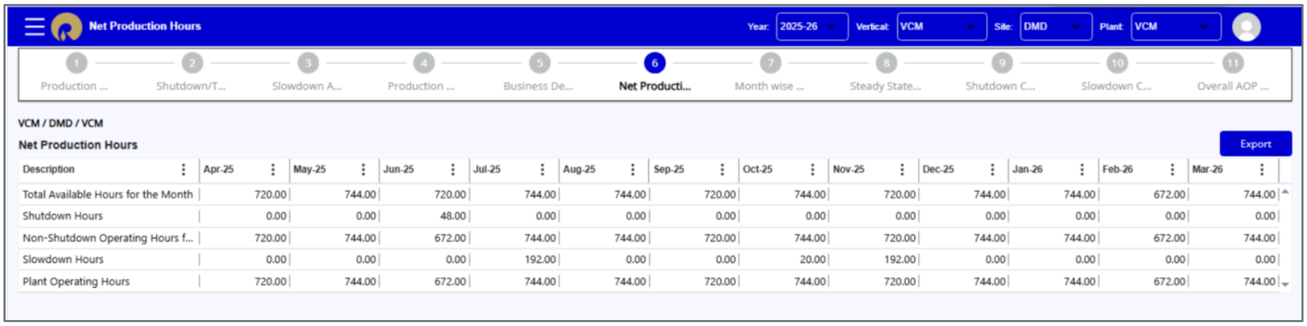
The user has to enter the SD-From and SD-To date and time to calculate the duration in hours, or SD-From and Duration(hrs) to calculate the SD-To date/time.

*The **'Add Item'** button on the Shutdown/Slowdown activities screen allows users to insert a new entry for shutdown/slowdown.*

The 'Save' button enables users to save newly added items as well as any modifications made to existing entries.

The screenshot below depicts the Net Production Hour details captured in the AOP digitized platform.

NOTE- *The below screen is read-only. The users cannot change the values*.



Description	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26
Total Available Hours for the Month	720.00	744.00	720.00	744.00	744.00	720.00	744.00	720.00	744.00	744.00	672.00	744.00
Shutdown Hours	0.00	0.00	48.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Non-Shutdown Operating Hours f...	720.00	744.00	672.00	744.00	744.00	720.00	744.00	720.00	744.00	744.00	672.00	744.00
Slowdown Hours	0.00	0.00	0.00	192.00	0.00	0.00	20.00	192.00	0.00	0.00	0.00	0.00
Plant Operating Hours	720.00	744.00	672.00	744.00	744.00	720.00	744.00	720.00	744.00	744.00	672.00	744.00

Figure 7 Digitized AOP screen for Net Production Hours

2.1.1.5 Month wise Production plan

Using the Business Demand, Production Target and Maintenance Plan (Operating Hours), the month-wise production quantity is derived.

The month-wise budget production for each product is calculated in MT and KT, the formula for which is mentioned below.

$$VCM\ (MT) = Proposed\ Operating\ Capacity * Operating\ Hours * (Business\ Demand/100)$$

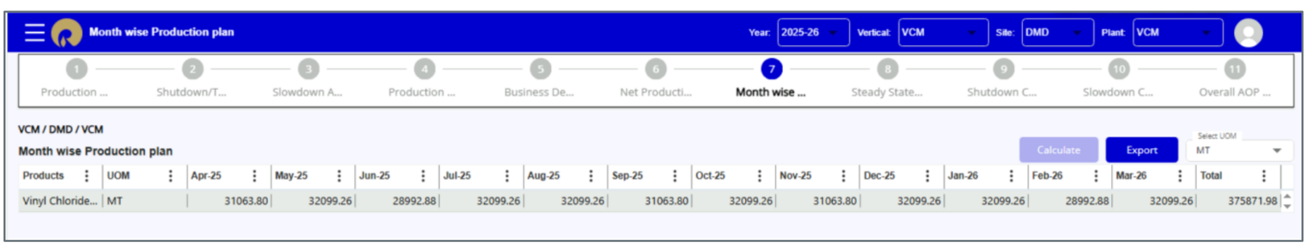
$$Operating\ hours\ = Total\ Available\ hours\ of\ the\ month\ - Shutdown/TA\ hours\ of\ the\ month$$

Hence, total budgeted production is,

$$Total\ Budgeted\ Production\ = \sum All\ the\ values\ for\ the\ month$$

The month-wise budget production is further used to derive the annual plan in KT, and this is part of the final report.

This information is captured in the AOP digitization portal as seen in the screenshot below.



Products	UOM	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Total
Vinyl Chloride...	MT	31063.80	32099.26	28992.88	32099.26	32099.26	31063.80	32099.26	31063.80	32099.26	32099.26	28992.88	32099.26	375871.98

Figure 8 Digitized AOP screen for Month wise Production Plan

The ***'Calculate'*** *button on this screen allows users to recalculate Production values when changes are made to production target, business demand, or when line items are added to shutdown activities.*

2.1.2 Norms

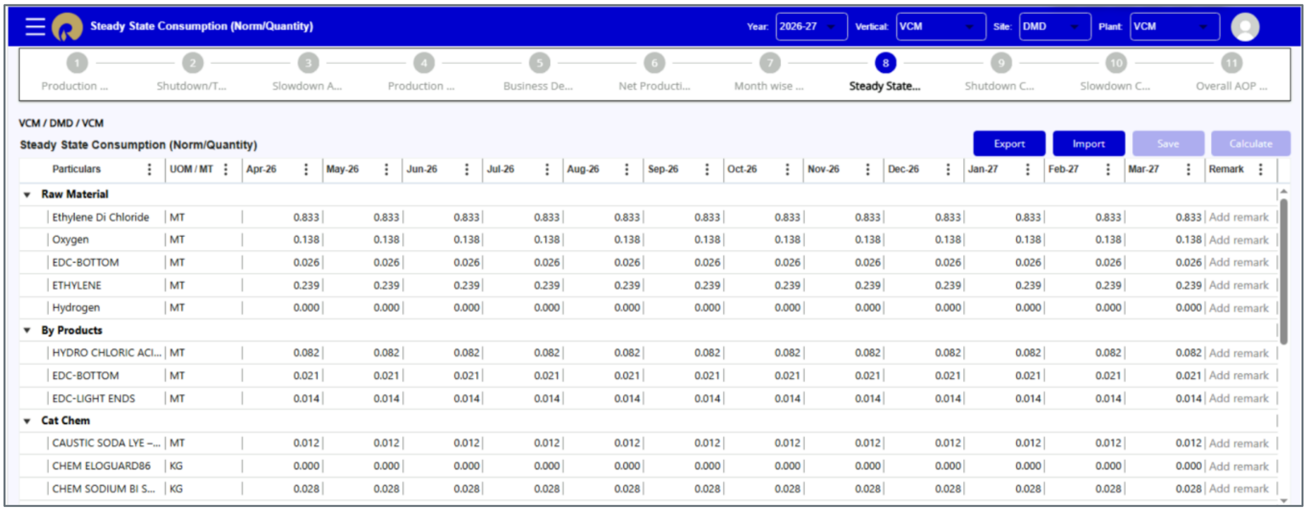
Below are the steps captured to derive the Norms value.

2.1.2.1 Steady State Norms

The Steady state norms screen serves as the basis to calculate steady state norms, allowing users to input essential data for consumption AOP calculations.

Steady state norms are calculated from SAP system by taking the average of available 02 or 03 consecutive monthly norms in last 05 years with average Monthly Production is greater than or equal to 1035 TPD.

The screenshot below highlights the details captured in the digitized AOP platform in the Steady state norms.



Particulars	UOM / MT	Apr-26	May-26	Jun-26	Jul-26	Aug-26	Sep-26	Oct-26	Nov-26	Dec-26	Jan-27	Feb-27	Mar-27	Remark
Raw Material														
Ethylene Di Chloride	MT	0.833	0.833	0.833	0.833	0.833	0.833	0.833	0.833	0.833	0.833	0.833	0.833	Add remark
Oxygen	MT	0.138	0.138	0.138	0.138	0.138	0.138	0.138	0.138	0.138	0.138	0.138	0.138	Add remark
EDC-BOTTOM	MT	0.026	0.026	0.026	0.026	0.026	0.026	0.026	0.026	0.026	0.026	0.026	0.026	Add remark
ETHYLENE	MT	0.239	0.239	0.239	0.239	0.239	0.239	0.239	0.239	0.239	0.239	0.239	0.239	Add remark
Hydrogen	MT	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	Add remark
By Products														
HYDRO CHLORIC ACI...	MT	0.082	0.082	0.082	0.082	0.082	0.082	0.082	0.082	0.082	0.082	0.082	0.082	Add remark
EDC-BOTTOM	MT	0.021	0.021	0.021	0.021	0.021	0.021	0.021	0.021	0.021	0.021	0.021	0.021	Add remark
EDC-LIGHT ENDS	MT	0.014	0.014	0.014	0.014	0.014	0.014	0.014	0.014	0.014	0.014	0.014	0.014	Add remark
Cat Chem														
CAUSTIC SODA LYE -...	MT	0.012	0.012	0.012	0.012	0.012	0.012	0.012	0.012	0.012	0.012	0.012	0.012	Add remark
CHEM ELOGUARD86	KG	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	Add remark
CHEM SODIUM BI S...	KG	0.028	0.028	0.028	0.028	0.028	0.028	0.028	0.028	0.028	0.028	0.028	0.028	Add remark

Figure 9 Digitized AOP screen for Steady state Norms

2.1.2.2 Shutdown Norms

Shutdown norms are recorded for the months in which shutdown activities are planned, as specified in the Shutdown/Turnaround Activities screen.

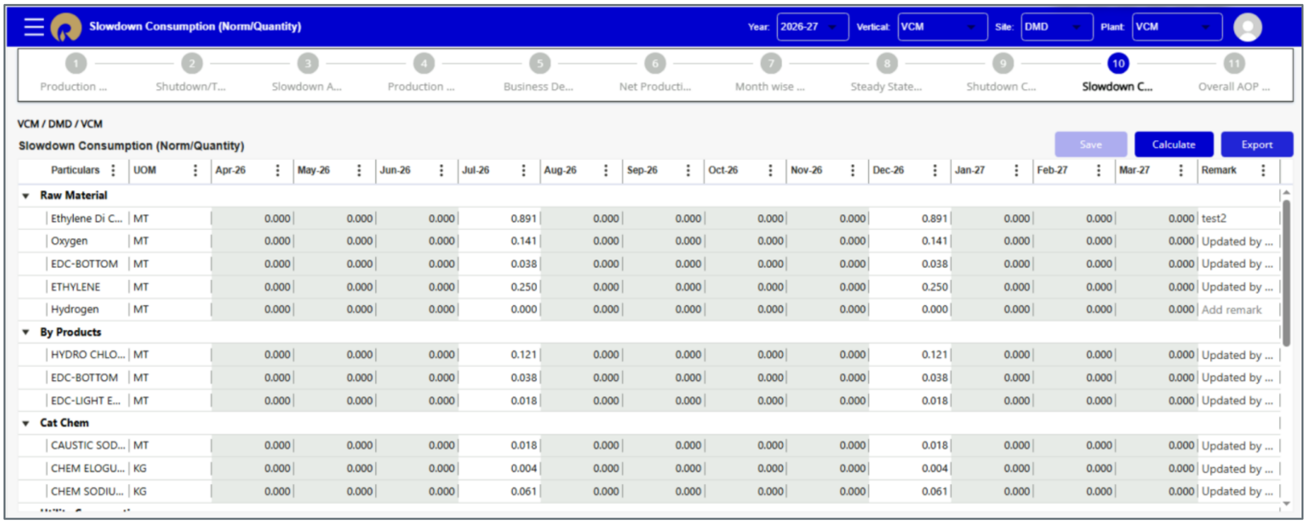
The shutdown quantity utilization data includes TA, asset maintenance linked shutdown/ slowdown activities, etc.

The use must enter the norms data for each line item.

2.1.2.3 Slowdown Norms

Slowdown norms are recorded for the months in which slowdown activities are planned, as specified in the Slowdown Activities screen.

The screenshot below highlights the details captured in the digitized AOP platform in the Slowdown norms.



Particulars	UOM	Apr-26	May-26	Jun-26	Jul-26	Aug-26	Sep-26	Oct-26	Nov-26	Dec-26	Jan-27	Feb-27	Mar-27	Remark
Raw Material														
Ethylene Di C...	MT	0.000	0.000	0.000	0.891	0.000	0.000	0.000	0.000	0.891	0.000	0.000	0.000	test2
Oxygen	MT	0.000	0.000	0.000	0.141	0.000	0.000	0.000	0.000	0.141	0.000	0.000	0.000	Updated by ...
EDC-BOTTOM	MT	0.000	0.000	0.000	0.038	0.000	0.000	0.000	0.000	0.038	0.000	0.000	0.000	Updated by ...
ETHYLENE	MT	0.000	0.000	0.000	0.250	0.000	0.000	0.000	0.000	0.250	0.000	0.000	0.000	Updated by ...
Hydrogen	MT	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	Add remark
By Products														
HYDRO CHLO...	MT	0.000	0.000	0.000	0.121	0.000	0.000	0.000	0.000	0.121	0.000	0.000	0.000	Updated by ...
EDC-BOTTOM	MT	0.000	0.000	0.000	0.038	0.000	0.000	0.000	0.000	0.038	0.000	0.000	0.000	Updated by ...
EDC-LIGHT E...	MT	0.000	0.000	0.000	0.018	0.000	0.000	0.000	0.000	0.018	0.000	0.000	0.000	Updated by ...
Cat Chem														
CAUSTIC SOD...	MT	0.000	0.000	0.000	0.018	0.000	0.000	0.000	0.000	0.018	0.000	0.000	0.000	Updated by ...
CHEM ELOGU...	KG	0.000	0.000	0.000	0.004	0.000	0.000	0.000	0.000	0.004	0.000	0.000	0.000	Updated by ...
CHEM SODIU...	KG	0.000	0.000	0.000	0.061	0.000	0.000	0.000	0.000	0.061	0.000	0.000	0.000	Updated by ...

Figure 10 Digitized AOP screen for Steady state Norms

2.1.2.4 Budget Norms/Consumption AOP

Using the values of Steady state norms and Shutdown Consumptions the final Consumption AOP value is calculated as below.

$$Consumption\ AOP\ for\ months\ without\ shutdown\ =\ Steady\ State\ Norms$$

$$Consumption\ AOP\ for\ months\ with\ shutdown\ =\ Steady\ State\ Norms + Shutdown\ Norms$$

3. AOP Reports

Following report will be implemented as part of digital AOP workflow.

- Annual AOP Cost
- Plant Production Summary
- Month wise Production Plan
- Month wise raw data
- Turnaround report
- Annual Production Plan
- Plant Contribution (T-21)
- Plant Contribution Summary (T-22)

4. Key Assumptions and Exemptions

The above scope of work should be read with the following assumptions made

1. Data Acceptance
2. DCT provided by DMD Site team